Date 2080/12/07 **Physics Practical Sheet** Experiment No 02

Class BSc 2nd year Group

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Shift Day Set

Object of the Experiment (Block Letters): TO CONSTRUCT AND VERIFY BASIC LOGIC GATES USING TTL (TRANSISTOR-TRANSISTOR LOGIC).

APPARATUS REQUIRED:

(i) Transistor (ii) Resistor (iii) Emitter
(iv) Wires (v) Base (vi) Multimeter
(vii) Collector (viii) Bread board (ix) Power Supply

THEORY:

Transistor-transistor logic (TTL) is a logic family built from bipolar junction transistor. Its name signifies that transistor performs both the logic function (the first "transistor") and the amplifying function (the second transistor).

Logic gates are electronic circuits which makes logic decision. They are building blocks of digital system. Function of logic gates are based on Boolean equation. There are two types of logic gate. They are simple or basic gates and compound logic gate.

(I) OR Gate:

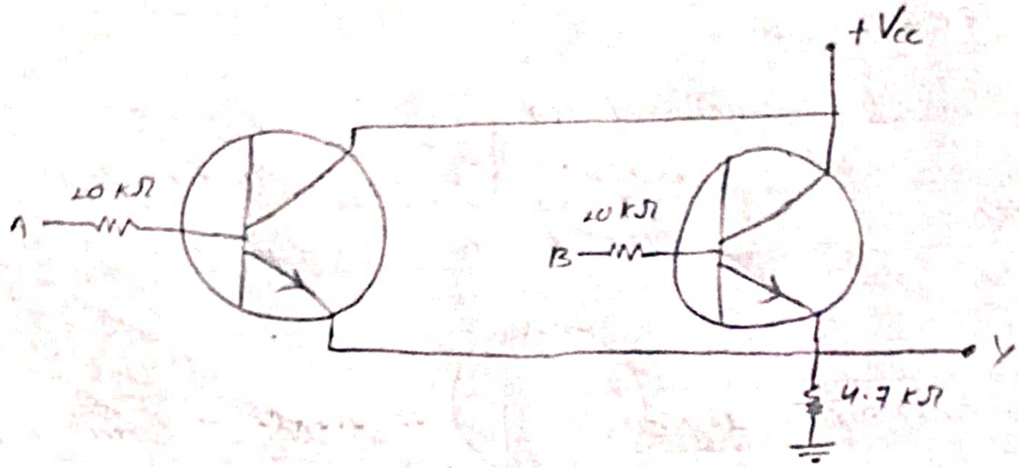
It has two or more inputs and single output. Its input is high if any or all input are high. Logic symbol, Boolean expression, truth table, practical circuit and realization circuit of two input OR gates are as follows:

The Boolean expression is,

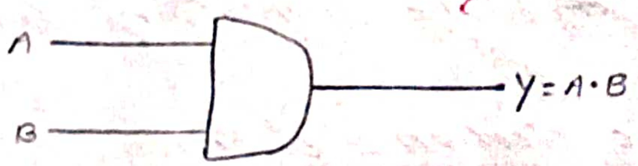
$$Y = A + B$$

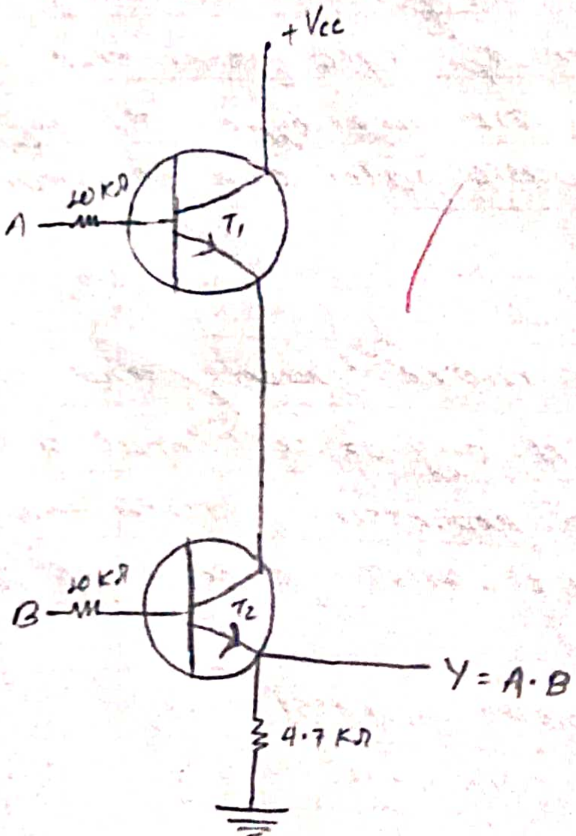
I OR GATE

Symbol



II AND GATE:





Truth Table

S.N	Inputs		Output
	A	B	$Y = A + B$
1.	0	0	0
2.	0	1	1
3.	1	0	1
4.	1	1	1

II) AND Gate:

It has two or more inputs and a single output. Its output is high if all inputs are high. Logic symbol, Boolean equation, truth table, practical circuit diagram of AND gates are as follows:

The Boolean expression is,

$$Y = A \cdot B$$

Truth Table

S.N.	Inputs		Output
	A	B	$Y = A \cdot B$
1.	0	0	0
2.	0	1	0
3.	1	0	0
4.	1	1	1

III) NOT Gate:

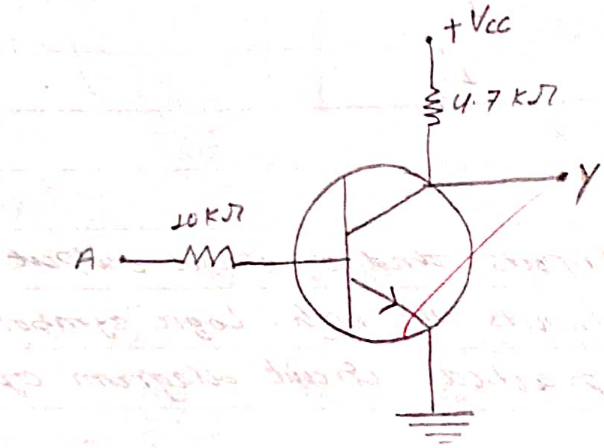
It has one input and one output. It reverse the nature of input. It converts true input to false and vice-versa.

The Boolean expression is,

$$Y = \bar{A}$$

i.e., $Y = \text{NOT } A$

III NOT GATE :



Truth Table

S.N.	Input	Output
	A	$Y=\bar{A}$
1.	0	1
2.	1	0

OBSERVATION :

① OR GATE

S.N.	Inputs		Output
	A	B	$Y = A + B$
1.	0	0	0
2.	[illegible]	3.99	3.33
3.	3.99	0	3.39
4.	3.99	3.99	3.42

(II) AND GATE :

S.N	Inputs		Output
	A	B	$Y = A \cdot B$
1.	0	0	0
2.	0	4	0
3.	4	0	0
4.	4	4	3.33

(III) NOT GATE :

S.N.	Input	Output
	A	$Y = \bar{A}$
1.	0	8.77
2.	9.16	0

RESULT:

From the above experiment, the function of simple logic gates are observed by using TTL.

CONCLUSION:

The basic logic gates is verified using transistor-transistor logic (TTL).

PRECAUTIONS:

(i) Common emitter base should be connected properly.
(ii) Connection of three transistor should be right.
(iii) Connection of two transistor should be right.
(iv) Wire should be connected properly.

12/1/6